B.Sc. Semester-V 2022
Computer Science
CC-12: Theory of Computation

Time: 3 hours **Full Marks: 60**

Answer question number 1 and any five from the rest

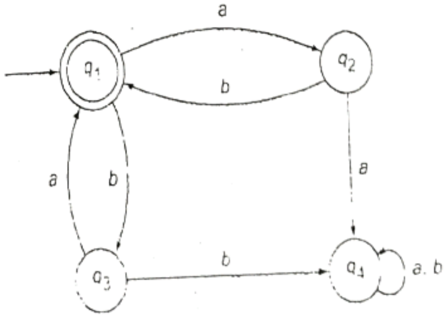
1 a) State Arden`s theorem.
b) How can you show that a grammar is ambiguous?
c) What is the advantage of PDA over FA?
d) What are the applications of pumping lemma for regular sets?
e) Distinguish between the phrases "the set is accepted by PDA by final state" and "the set is accepted by PDA by null store (or empty store)". [2×5]

2 a) Construct a DFA accepting all strings over $\{0, 1\}$ such that the number of 1`s in w is $3 \bmod 4$.
b) Construct a Mealy machine which is equivalent to the following Moore machine: [5+5]

Present State	Next State		Output
	a=0	a=1	
q_0	q_1	q_2	0
q_1	q_3	q_2	0
q_2	q_2	q_1	1
q_3	q_0	q_3	1

3 a) Find the language generated by the following grammars:
(i) $S \rightarrow 0S1 \mid 0A1 \quad A \rightarrow 1A \mid 1$
(ii) $S \rightarrow 0A \mid 1S \mid 1 \mid 0 \quad A \rightarrow 1A \mid 1S \mid 1$
b) Construct the grammar, accepting each of the following sets:
(i) The set of all strings over $\{0,1\}$ consisting of an equal number of 0's and 1's.
(ii) $\{0^n 1^m 0^m 1^n \mid m, n \geq 1\}$ [4+6]
(iii) $\{0^n 1^{2n} \mid n \geq 1\}$

4 a) What is a regular expression? State any four identities for regular expressions.
b) Prove that the finite automaton whose transition diagram is as shown in the figure below accepts the set of all strings over the alphabet $\{a, b\}$ with an equal number of a's and b's, such that each prefix has at most one more a than the b's and at most one more b than the a's. [4+6]



5 a) Construct a DFA with reduced states equivalent to the regular expression
$(00 + 11)\, 0^*1 + 01.$ [5+5]
b) Show that $L = \{0^i 1^i \mid i \geq 1\}$ is not regular.

6 a) Given a grammar free from unit productions and null productions, how can you simplify it?

b) How can you reduce any context free grammars into its equivalent Chomsky Normal Form? [5+5]

7 a) Design a Turing machine M to recognize the language $\{1^n 2^n 3^n \mid n \geq 1\}$.

b) Check whether the strings 112233 and 1233 are accepted by the Turing machine designed above. [6+4]

8 Write short notes on any two of the followings:

a) Ambiguous grammar

b) Removal of Λ-moves

c) Advantages and disadvantages of PDA over FA [5+5]
